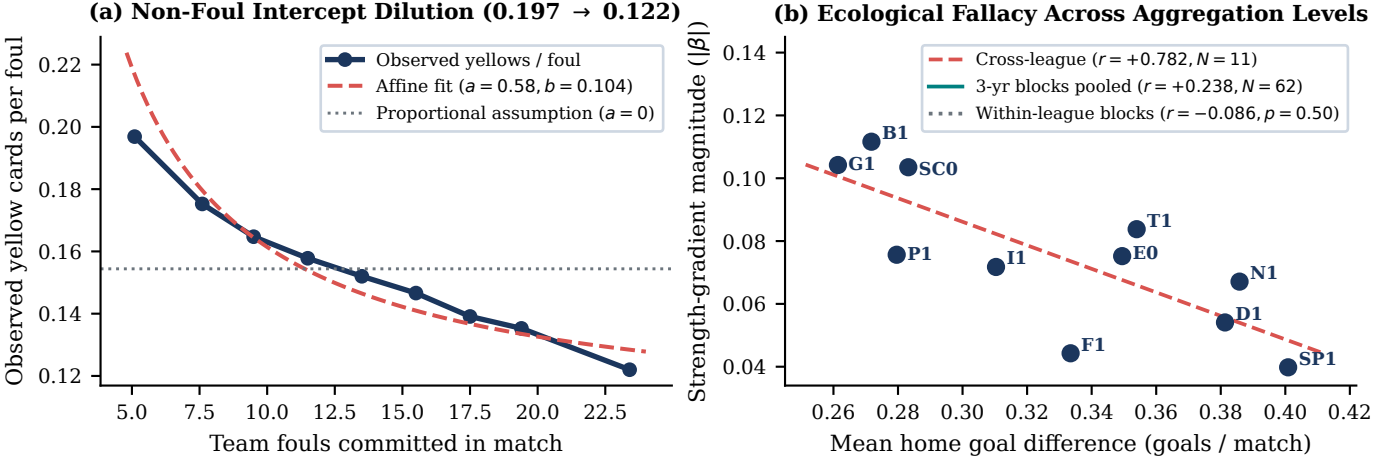
(a) Non-Foul Intercept Dilution (0.197 → 0.122)
Observed yellows / foul
Affine fit (a = 0.58, b = 0.104)
Proportional assumption (a = 0)
Observed yellow cards per foul
0.22
0.20
0.18
0.16
0.14
0.12
5.0
7.5
10.0
12.5
15.0
17.5
20.0
22.5
Team fouls committed in match
(b) Ecological Fallacy Across Aggregation Levels
Cross-league (r = +0.782, N = 11)
3-yr blocks pooled (r = +0.238, N = 62)
Within-league blocks (r = −0.086, p = 0.50)
Strength-gradient magnitude (|β|)
0.14
0.12
0.10
0.08
0.06
0.04
B1
G1
SC0
T1
P1
I1
E0
N1
D1
F1
SP1
0.26
0.28
0.30
0.32
0.34
0.36
0.38
0.40
0.42
Mean home goal difference (goals / match)